Project Title: MediTrust

UN- SDG initiative

Central India Hackathon 3.0

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Domain: Health and Care

Problem Statement

Access to timely, affordable, trustworthy healthcare is a major challenge in rural and underserved areas. Patients struggle to assess condition severity, connect with verified doctors quickly, and afford medicines and diagnostics. Current platforms overload doctors with minor cases or lack trust, verification, and cost optimization.

AI-Assisted Healthcare Platform

Build an AI-powered platform with an intelligent triage system to evaluate condition severity. For non-severe cases, provide guided AI assistance and recommendations; for critical cases, instantly escalate to the best-matched, verified doctor.

Key Features:

- Video/audio/chat consultations and online/offline appointment booking.
- Lab/diagnostic bookings.
- Doctor verification via government registries.
- Cost-effective generics (doctor-approved only).
- Transparent ratings/reviews.

Ensure affordability, scalability for low-connectivity areas, patient safety, data privacy, and ethical AI.

Proposed Solution

“From symptoms to solutions—fast, personalized care powered by AI and expert doctors.”

1. Quick Symptom Assessment

Problem: People don’t know if their symptoms are serious.

Solution: AI triage quickly evaluates symptoms and advises whether to see a doctor, try home care, or monitor.

2. Timely Doctor Access

Problem: Delays in getting medical attention.

Solution: Automatically connects patients to doctors for severe cases or appointments, reducing waiting time.

3. Personalized Care Guidance

Problem: Generic health advice doesn’t fit everyone.

Solution: Offers tailored guidance based on age, medical history, and symptom severity.

4. Efficient Healthcare Management

Problem: Doctors’ time is wasted on minor issues.

Solution: AI filters mild cases, letting doctors focus on critical patients.

5. Health Monitoring & Records

Problem: Patients struggle to keep track of health data.

Solution: Stores health records, tracks symptoms, medications, and doctor visits in one app.

6. Preventive Health Insights

Problem: People often neglect preventive care.

Solution: App suggests preventive measures, lifestyle tips, and reminders for checkups.

7. Data-Driven Decisions

Problem: Lack of insights into health trends.

Solution: Aggregated, anonymized data can help identify patterns, risks, and early warnings.

INNOVATIVE FACTOR

“From symptoms to solutions—fast, personalized care powered by AI and expert doctors.”

- **AI-Driven Triage:** Our app connects patients to an AI first, which can assess the severity of the condition before routing them to a doctor. This reduces unnecessary doctor visits and optimizes healthcare resources.
- **Automatic Escalation:** If the AI detects a severe condition, the system automatically connects the patient to a doctor. This ensures timely intervention, which is often missing in traditional telemedicine apps.
- **Personalized Care Pathways:** Based on symptoms, history, and AI analysis, the app can suggest next steps, preventive care, or home remedies tailored to the patient—making healthcare more personalized.
- **Resource Optimization:** By filtering mild cases through AI, doctors’ time is freed for critical cases, improving efficiency in clinics or telehealth setups.
- **Data-Driven Insights:** Continuous collection of anonymized health data could help in predicting outbreaks, identifying health trends, and improving overall healthcare delivery.
- **User Convenience & Engagement:** Combining AI guidance with seamless doctor access provides a smoother and faster patient experience compared to standard apps that only allow appointments.

Implementation Overview

Tech Used:-

- MongoDB - for Database
- Expressjs/Nodejs- for Backend
- React - for Frontend
- Python for AI integration

React UI

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Node.js/Express.js API

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AI Service (API/Python)

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Decision Engine

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Doctor / Patient

Impact & Feasibility

Impact

1. **Improved Patient Outcomes:** Faster assessments and timely doctor connections can prevent complications and save lives.
2. **Reduced Healthcare Burden:** AI filtering of minor cases reduces unnecessary hospital visits, freeing up doctors for critical cases.
3. **Personalized Care:** Tailored guidance increases adherence to treatment and preventive measures.
4. **Data-Driven Insights:** Aggregated anonymized health data can inform public health strategies and early warnings.
5. **Enhanced Convenience:** Patients get healthcare guidance anytime, anywhere, reducing stress and travel time.

Feasibility

1. **Technology Readiness:** AI symptom checkers and telemedicine platforms already exist; combining them is realistic.
2. **Integration with Healthcare Providers:** APIs and telehealth networks make doctor connection feasible.
3. **User Adoption:** Mobile-first, easy-to-use interface encourages widespread use.
4. **Data Security:** Secure cloud storage and privacy protocols ensure compliance with healthcare standards.
5. **Scalability:** The platform can start regionally and scale nationwide or globally.

Revenue Model & Future Scope

Revenue Model

1. Freemium Model

- Free: AI symptom check, basic health guidance
- Premium: Instant doctor consultations
- Priority access
- Detailed health reports

2. Pay-Per-Consultation

- Users pay only when they connect with a doctor.
- Platform takes a commission per consultation

3. Subscription Plans

- Monthly / yearly plans for:
  - Families
  - Chronic patients
  - Senior citizens
  - Includes reminders, follow-ups, and priority care.

Future Scope

1. Advanced AI Models

- Improved severity prediction
- Personalized treatment suggestions
- Predictive health alerts

2. Mobile App Expansion

- Android & iOS apps
- Wearable integration (heart rate, steps, SpO_2)

3. Multi-Language & Voice AI

- Regional language support
- Voice-based symptom input (for rural users)

4. Remote Monitoring

- IoT medical devices
- Continuous vitals tracking
- Alerts for abnormal readings

5. Integration with Health Systems

- Government health programs
- Insurance claim integration
- Electronic Health Records (EHR)

Conclusion

This app positions itself as a comprehensive health companion, prioritizing user safety, accessibility, and innovation. By addressing common pain points like
doctor availability and medicine authenticity, it has the potential to revolutionize healthcare delivery in India. Feedback and iterations will be key to its success.